Linear reactivity model

NE 630 | Nuclear Reactor Theory

24
LESSON

Read: Linear Reactivity Model supplement; review Lesson 23.

How do batch refueling strategies change cycle length and discharge burnup?

Objective: Estimate cycle lengths and discharge burnups using the linear reactivity model.

1 Review: reactivity and depletion

For constant total flux and one-group cross sections, constant $N^{^{238}\mathrm{U}}$, and $N^{^{239}\mathrm{Pu}}(0) = 0$, Lesson 23 gives

$$N^{^{235}\mathrm{U}}(t) = N^{^{235}\mathrm{U}}(0)e^{-\bar{\sigma}_a^{^{235}\mathrm{U}}\phi t},$$
$$N^{^{238}\mathrm{U}}(t) = N^{^{238}\mathrm{U}}(0),$$
$$N^{^{239}\mathrm{Pu}}(t) = \frac{\bar{\sigma}_\gamma^{^{238}\mathrm{U}}}{\bar{\sigma}_a^{^{239}\mathrm{Pu}}} N^{^{238}\mathrm{U}}(0)\left[1 - e^{-\bar{\sigma}_a^{^{239}\mathrm{Pu}}\phi t}\right].$$

Neglecting control and fission-product absorption,

$$k(t) = \frac{\left[\bar{\nu}_T^{^{235}\mathrm{U}}\bar{\sigma}_{fT}^{^{235}\mathrm{U}}N^{^{235}\mathrm{U}}(t) + \bar{\nu}_T^{^{239}\mathrm{Pu}}\bar{\sigma}_{fT}^{^{239}\mathrm{Pu}}N^{^{239}\mathrm{Pu}}(t)\right]\varepsilon p P_{\mathrm{NL}}}{\bar{\sigma}_{aT}^{^{235}\mathrm{U}}N^{^{235}\mathrm{U}}(t) + \bar{\sigma}_{aT}^{^{239}\mathrm{Pu}}N^{^{239}\mathrm{Pu}}(t) + \Sigma_{aT}^{^{238}\mathrm{U}} + \varsigma\frac{V_m}{V_f}\Sigma_{aT}^m}, \quad \text{(FNRP 10.39*)}$$

with $\rho = (k-1)/k$. The asterisk makes thermal-group (T) cross sections and nuclide-specific neutron yields per fission explicit.

2 Let's get some rough PWR numbers

Let's do some back-of-the-envelope calculations to get rough (one- or two-significant-figure) numbers for a typical PWR. Use the rounded $V_F = 10^7$ cm^3 in parts 2–4; burnup is per initial uranium mass.

1. Fuel volume. A 3000 $\mathrm{MW_{th}}$ PWR has about 50,000 rods, each roughly 3 m high. With $r \simeq 0.5$ cm,

$$V_F \simeq (5\times10^4)\pi r^2 h \simeq _____.$$

2. Flux. If $\bar{\Sigma}_f = 0.05$ cm^{-1} and $\kappa \simeq 3.2\times10^{-11}$ J/fission, then

$$\phi = \frac{P}{V_F\bar{\Sigma}_f\kappa} \simeq _____.$$

3. Fuel mass. With $\rho_{\mathrm{UO_2}} = 10$ g/cm^3,

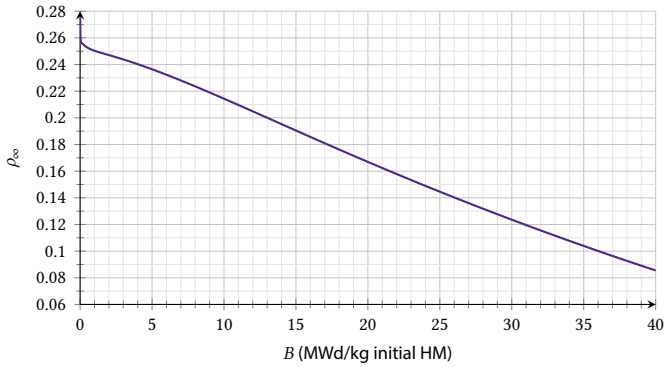
$$m_{\mathrm{UO_2}} = \rho_{\mathrm{UO_2}}V_F \simeq _____,$$
$$m_{\mathrm{U}} \simeq \frac{238}{270}m_{\mathrm{UO_2}} \simeq _____.$$

4. Burnup. For five full-power years (not calendar years),

$$B_{1\,\mathrm{d}} \simeq \frac{3000\ \mathrm{MWd}}{8.8\times10^4\ \mathrm{kg}} \simeq _____,$$
$$B_{5\,\mathrm{yr}} \simeq 5(365)B_{1\,\mathrm{d}} \simeq _____.$$

3 Use a more realistic reactivity curve

Use the fixed-power curve from `lesson_24.ipynb`: 4% enriched UO_2 at 10.4 g/cm^3, 300 W/cm^3 fuel, and the expanded chain with Xe/Sm. The prescribed-spectrum rate ratio uses $P_{\mathrm{NL}} = 1$; leakage is treated separately below.



By eye, extend the post-xenon trend back to $B = 0$. For this exercise, take the line through $(0, 0.25)$ and $(35, 0.10)$ exactly. Estimate its slope and intercept in Step 5. The curve has no zero crossing in the plotted range.

4 State the three LRM assumptions

Assumption 1: identical batches

Use N equal-mass batches of initially identical fuel; each gains burnup B_c per cycle. For $N = 3$, refueling replaces one third of the core.

At **BOC**, batch burnups are $0, B_c, 2B_c$; at **EOC**, they are $B_c, 2B_c, 3B_c$. The oldest batch is discharged. In this model, EOC exhausts available excess reactivity.

B_c is the **cycle burnup**; $B_d = NB_c$ is the **discharge burnup**. These are burnups, not elapsed times.

Assumption 2: linear fuel reactivity

$$\boxed{\rho_\infty(B) = \rho_0 - aB.}$$

The subscript ∞ denotes infinite-medium fuel reactivity. Treat the constant leakage penalty ρ_L separately; $a > 0$ is the loss per unit burnup.

Assumption 3: weighted core reactivity

$$\boxed{\rho_{\mathrm{core}} = \sum_{i=1}^{N} w_i\rho_i - \rho_L}, \qquad \boxed{\sum_{i=1}^{N} w_i = 1}.$$

Here $\rho_i = \rho_\infty(B_i)$; w_i are fixed weights. We use $w_i = 1/N$. Adjustable control offsets positive ρ_{core} during operation.

Lesson 24 | Batches, cycle length, and discharge burnup

5 Fit a linear fuel-reactivity model

Use the eyeballed points $(B_1, \rho_1) = (0, 0.25)$ and $(B_2, \rho_2) = (35, 0.10)$ to approximate the post-xenon trend:

$$\frac{\Delta\rho}{\Delta B} = \frac{0.10 - 0.25}{35 - 0} = _____,$$

$$a = -\frac{\Delta\rho}{\Delta B} = _____,$$

$$\rho_0 = \rho_1 + aB_1 = _____.$$

Reactivity is dimensionless; a has units $(\text{MWd/kg})^{-1}$. MWd/kg initial HM is numerically equal to GWd/MTU for the initial UO_2 fuel. The intercept approximates the post-xenon trend extended back to $B = 0$, rather than the fresh-fuel startup value.
Use this same two-point approximation for the numerical checks below.

6 Test a proposed two-batch cycle

Take $N = 2$, $\rho_L = 0.05$, and suppose $B_c = 10$ GWd/MTU.
At BOC, the two batches have burnups 0 and B_c:

$$\rho_{\text{core}}^{\text{BOC}} = \frac{1}{2}\left[\rho_\infty(0) + \rho_\infty(B_c)\right] - \rho_L$$

$$= _____.$$

At EOC, the same batches have burnups B_c and $2B_c$:

$$\rho_{\text{core}}^{\text{EOC}} = \frac{1}{2}\left[\rho_\infty(B_c) + \rho_\infty(2B_c)\right] - \rho_L$$

$$\simeq _____.$$

Because $\rho_{\text{core}}^{\text{EOC}} > 0$, this model permits a longer cycle. Actual cycles also face thermal, fuel-integrity, and operating constraints.

Before continuing

Sketch the two batch burnups at BOC and EOC. Which batch is discharged? Which one is replaced by fresh fuel?

7 Choose B_c so EOC is critical

For the two-batch core at EOC,

$$\bar{\rho}_\infty^{\text{EOC}} = \frac{1}{2}\left[(\rho_0 - aB_c) + (\rho_0 - 2aB_c)\right]$$

$$= \rho_0 - \frac{3}{2}aB_c.$$

Set $\rho_{\text{core}}^{\text{EOC}} = 0$:

$$0 = \rho_0 - \frac{3}{2}aB_c - \rho_L, \qquad \boxed{B_c = \frac{2(\rho_0 - \rho_L)}{3a}}.$$

With the eyeballed two-point model,

$$B_c \simeq _____,$$

$$B_d = 2B_c \simeq _____.$$

For a one-batch core,

$$B_c^{(1)} = B_d^{(1)} = \frac{\rho_0 - \rho_L}{a} \simeq _____.$$

Range check: These limiting burnups extrapolate beyond the 40 MWd/kg fit range; they are algebraic LRM results, not recommended fuel-use limits.
Thus, the two-batch core has a shorter cycle, $B_c^{(2)} = \frac{2}{3}B_c^{(1)}$, but a higher discharge burnup, $B_d^{(2)} = \frac{4}{3}B_d^{(1)}$–about 33% higher.

8 Generalize to an N-batch core

At EOC, the batch reactivities are

$$\rho_1 = \rho_0 - aB_c,$$
$$\rho_2 = \rho_0 - 2aB_c,$$
$$\vdots$$
$$\rho_N = \rho_0 - NaB_c.$$

For equal batch weights,

$$0 = \rho_{\text{core}}^{\text{EOC}}$$

$$= \frac{1}{N}\left[N\rho_0 - aB_c(1 + 2 + \cdots + N)\right] - \rho_L.$$

Using $1 + 2 + \cdots + N = N(N+1)/2$,

$$\boxed{B_c^{(N)} = \frac{2(\rho_0 - \rho_L)}{a(N+1)}}, \qquad \boxed{B_d^{(N)} = NB_c^{(N)}}.$$

For the same ρ_0, a, and constant ρ_L for all N (with $\rho_0 > \rho_L$),

$$\boxed{B_c^{(N)} = \frac{2}{N+1}B_c^{(1)}}, \qquad \boxed{B_d^{(N)} = \frac{2N}{N+1}B_d^{(1)}}.$$

What the linear reactivity model says

Dividing a core into more equal batches makes each individual cycle shorter, but lets the discharged fuel remain in the core for more cycles and reach a larger burnup. The model is intentionally simple: it assumes an equilibrium cycle, equal batch burnup increments, linear fuel reactivity, fixed batch weights, and a constant leakage penalty.